APP3: School Scheduling

Algorithmic Problem Solving

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Source: https://imgflip.com/memetemplate/Boardroom-Meeting-Suggestion

Learning objectives:

- Understand the maximum flow and the minimum cut problems.
- Learn about approximation algorithms.

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I. APP3: School Scheduling

I.1. APP Objectives

After this APP3, you will be capable of:

- working with graphs to solve difficult problems;
- create and analyze approximation algorithms.

I.2. Session Organization

Follow the guidelines of the previous project.

I.3. The Problem

Situation — problem

Your childhood school asked you to help them schedule their courses this year. Like many schools, they too have some financial difficulties. To mitigate this issue, the director has decided to use as few classrooms as possible during the year to cut down on air conditioning (cooling and heating) expenses.

There are N courses in the school and each course is taught every day. You have received the starting time A_i and finishing time B_i (inclusive) of each course i ($1 \leq i \leq N$). The courses have different number of students S_i ($1 \leq i \leq N$). There are no students who are registered to overlapping courses. Each classroom has a fixed capacity to hold M students. They teach many different courses at the school, so the teachers need some preparation time between courses to clean up and get the necessary equipment ready. Depending on the order of the courses, this preparation time might differ (cleaning up after a chemistry class takes much longer than arranging the books after literature). In your supplied documents, there is a matrix which specifies the preparation time P_{ij} needed between courses i and j.

Your job is to help your school to determine the minimum number of rooms need for the courses.

Input. The input starts with two integers: N ($1 \leq N \leq 100$), the number of courses, and M ($1 \leq M \leq 10000$), the capacity of one classroom. Next, there are N lines containing three integers: A_i, B_i ($0 \leq A_i \leq B_i \leq 10000000$), and S_i ($1 \leq S_i \leq 10000$), the starting and finishing times, and the number of students of course i. The ending time is **inclusive**, meaning that if a course finishes at unit time 10, the cleaning can only start at unit time 11. Afterwards, there are N lines detailing the preparation time matrix, where the i-th row contains N integers P_{ij} ($1 \leq i \leq N, 1 \leq j \leq N, 0 \leq P_{ij} \leq 10000000, P_{ii} = 0$). In other words, the row's index corresponds to the course which has finished, and the column's index to the course which should be started afterwards.

Output. An integer indicating the number of classrooms the school needs to operate.

CC BY SA

Situation — problem

Examples:

	Instance 1	**Instance 2**	**Instance 3**
Input	1 5 1 60 12 0	4 1 1 100 10 50 130 3 150 200 15 80 170 7 0 2 3 4 5 0 7 8 9 10 0 12 13 14 15 0	2 1 1 10 1 12 20 1 0 2 5 0
Output	3	22	2

I.4. Resources

This APP will cover basic knowledge on the maximum flow problem, the minimum cut problem, and approximation algorithms. It is required that you read the notes given at the address below.

- Maximum Flow Part1
- Maximum Flow Part2
- Minimum Cut

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